



Research Article

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Optimizing Cybersecurity Threat Detection: The Power of English for Specific Purposes (ESP) Proficiency

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Abstract

This study investigates the function of English for Specific Purposes (ESP) training in the capacity of the student in the identification and analysis of cybersecurity threats. Using the quasi-experimental mixed-methods approach, the data were collected from the participation of 100 participants consisting of 50 ESP-trained software engineering students and 50 non-ESP-trained computer engineering students. The study assessed ESP-related grammar, syntax, vocabulary, word formation, and contextual usage through a language proficiency test, alongside a survey on students' perceptions of ESP training's effectiveness in cybersecurity analysis. The respondents also engaged in interactions using cybersecurity texts and the TryHackMe platform in cyberattack reportage and analysis. Results demonstrated significant improvement among ESP-trained participants in threat analysis, technical reportage, and analytic reasoning. Gender-based differences were observed, as male participants were more assertive in terms of the risk assessment, while female participants were more precise in the documentation. Digital platforms were also observed to have the critical function in the facilitation of ESP learning and cybersecurity training. These results suggest towards the demand for the incorporation of ESP into cybersecurity education in bridging the linguistic-technical skills gap towards enhanced preparation in actual cyber threat identification and management. Future research should also explore the long-term implication of ESP training towards professional practice in cybersecurity and explore adaptive approach towards the acquisition of the specific field.

Keywords: English for Specific Purposes (ESP), cybersecurity threats, specialized vocabulary, grammar proficiency, contextual understanding

1. Introduction

The growing sophistication of cyber threats requires the creation of new approaches to enhancing the detection of cybersecurity threats, especially in the protection of sensitive information and infrastructure. The main approach includes the application of threat intelligence that utilizes strategic, tactical, and operational models in detecting highly evasive threats that can evade traditional security controls (Cyber2yr2020 Task Group, 2020). Against the backdrop of accelerating rates and levels of cyber-attacks, organizations are under pressure to enhance detection mechanisms to reduce vulnerabilities (Borky & Bradley, 2018). In such a scenario, language competency—English

for Specific Purposes (ESP) being no exception—as a key driver in effective cybersecurity practice has taken centerstage (Brown, 2024).

ESP, being a professional language learning approach, provides professionals with technical lexis and situational fluency necessary to operate in sophisticated cybersecurity contexts (Denizhalil, 2024). By focusing on use, ESP fills the gap between linguistic competence and threat analysis needs, thus allowing practitioners to explain technical ideas with clarity (Cybersecurity and Infrastructure Security Agency, n.d.). It is a critical skill for working on international teams and understanding international threat advisories, which are commonly issued in English (Johnson, 2018). Bell, D. (2001) also mentions the centrality of communication in cybersecurity due to the information age and how it has promoted the development of cybercultures, pointing out how workers now need to be able to function in an environment of cybersecurity that is globalized with linguistic and cultural adaptability.

In spite of this, ESP and cybersecurity convergence is confronted by over-reliance on specialized terminology that excludes laypersons, with potential for miscommunication and ineffective risk management (Yoon & Chang, 2021). Such communication gaps aggravate weaknesses because human mistakes are a leading factor in data breaches (Wagner & Alharthi, 2024). Closing such gaps requires pedagogical strategies that include ESP instruction supplemented by cybersecurity learning in a way that elevates communicative and technical abilities (Rahman, 2015). American Library Association (1989) focuses on information literacy in the ability to keep pace with the digital world through the note that people will be required to possess fundamental information literacy skills in an ever-evolving technologically founded environment.

Beatty, K. (2003) synthesizes this with the necessity of targeted language instruction for professional areas, such as cybersecurity, where workers must handle very technical terminology and global standards. Beatty (2003) emphasizes that effective ESP courses facilitate the understanding of complex technical concepts, and thereby, enhancement of professional competence, as well as organizational effectiveness.

The globalized nature of cybersecurity also underscores the need to apply ESP skills by experts. Threats have no borders, so the ability to read and write international threat intelligence reports is a goldmine. Effective communication not only aids improved threat identification but also enables cohesive responses across varied stakeholders, from technical teams to senior management (Asad, 2025). For instance, the use of plain language can enable inclusivity to engage more individuals in security processes (Cyber Management Alliance, 2024). Moreover, person-centered cybersecurity practices—emphasizing active listening training and concise documentation—have been successful in reducing errors that are caused by miscommunication. This practice augments research that focuses on interpersonal skills in addition to technical skills in cybersecurity training programs (Wood, n.d.).

Inclusion of ESP in cybersecurity systems also enhances the utilization of sophisticated detection methods. ESP practitioners can view security systems holistically, identify out-of-pattern anomalies, and incorporate threats into high-risk operational views (Matveeva & Fedotkina, 2021). It becomes particularly pertinent in the context of adaptive forms of threats such as zero-day exploit, beyond the capability of the majority of signature-based solutions (Cybersecurity and Infrastructure Security Agency, n.d.). Furthermore, ESP capabilities help to bring threat intelligence together so that companies can anticipate attacks rather than reacting.

To achieve maximum potential of ESP for learning about cybersecurity, pedagogical innovations are essential. The inclusion of case studies from actual environments in curricula allows students to place risks like ransomware and phishing attacks, thus creating analysis and decision-making abilities (Cai, 2018). Hands-on experience comes alongside tailored curricula mapping language acquisition with industry-specific situations for application within students' careers (Cyber Management Alliance, 2020). For example, case studies of multinational companies identify different strategies for defense against supply chain attacks, which enrich students' lists of problems (DigitalDefynd, 2024). A customized curriculum also addresses the language requirements of cybersecurity work, for

instance, drafting incident reports or explaining vulnerabilities to non-technical stakeholders (ABC Horizon, n.d.).

No less critical is coordination between technical and soft skills in ESP classes. Whereas technical classes focus on threat analysis language syntax and semantics, soft skills such as moral decision-making and group communication equip students with cooperation skills to integrate into collaborative interdisciplines (Armstrong, 2025). Practical exercises such as incident response simulation promote flexibility as well as capacity to operate evidently under stress (Beuran et al., 2019). In addition, technology-based learning environments, such as virtual laboratories and threat simulation systems, involve learners in authentic environments, connecting theory to practice.

Constant feedback cycles further refine ESP courses to conform to emerging industry standards (Kolomiets & Konoplenko, 2015). Ongoing monitoring of students' linguistic and technical progress enables instructors to bridge gaps in knowledge, particularly in idiomatic learning and syntax of cybersecurity reports. The repetitive process not only enhances personal competency but also organizational resistance against emerging threats.

Briefly, incorporating ESP into cyber security training is a revolutionary way of closing the linguistic-technical divide. By equipping professionals with the double skill set of advanced threat analysis and communication, organizations are able to build robust defenses against an ever-changing danger landscape. Such a double skill set is central to endeavoring overall readiness in cyber security in the ever-more globalized digital landscape.

To examine the effect of English for Specific Purposes (ESP) training on cybersecurity threat detection and technical communication, this paper is structured as follows: the research methodology used to assess the use of ESP in the context of training for cybersecurity threat detection is presented in the following section, with an emphasis on the research design, participant recruitment, and data collection instruments. The subsequent section offers results of the quasi-experimental mixed-methods study, both in terms of quantitative language proficiency as well as threat report evaluation data and qualitative responses obtained from instructors and students. Discussion interprets the findings, exploring the means through which ESP training influences technical communications skills, cross-cultural competence, and operational performance in cybersecurity operations. Finally, the paper ends with a summary of the key findings, recommendations for innovative curriculum for teaching cybersecurity, future research suggestions and recommendations, and an extensive list of references for scholarly backing of this research.

2. Methodology

2.1 Research Design

The research design implemented a quasi-experimental mixed-methods approach to evaluate how students perform when trained with English for Specific Purposes (ESP) in cybersecurity threat identification and analysis. Two naturally developed groups consisting of ESP-trained students and non-ESP-trained students from University Metropolitan Tirana participated in the quasi-experimental study instead of random allocation. When implementing this research, the researchers assessed the training effectiveness by using both quantitative results from language proficiency tests and threat analysis scores together with qualitative survey responses and observations of language use under real-world conditions. This combination of quantitative and qualitative data methods provided a deep examination of how the ESP program assists students in technical communication and analytic thinking for cybersecurity work. The research design sought to achieve both quantitative comparison and qualitative analysis of language abilities for authentic cybersecurity tasks.

2.2 Participants

A total of 100 participants divided equally between two separate groups represented the study sample.

- Group 1 – ESP-Trained Software Engineering Students (n = 50): The students received extensive training in English for Specific Purposes (ESP) education that specifically focused on cybersecurity during their studies. The training included grammar, syntax teaching along with technical vocabulary and word creation together with contextual language use for cybersecurity applications.
- Group 2 – Non-ESP-Trained Computer Engineering Students (n = 50): The participants in this group show basic competence with English yet they lack specialized training in cyber security through English for Specific Purposes (ESP) programs.

2.3 Materials

2.3.1 Cybersecurity Texts and Reports

The participants were given real cybersecurity reports, i.e., threat reports, incident reports, and analytic reports. These reports included technical terms and contained suitable grammar and syntactic structures as pertinent to cybersecurity.

2.3.2 Threat Simulation Platform – TryHackMe

Students were working on TryHackMe, an online practice environment for cybersecurity. The site provided virtual cybersecurity practice exercises comprised of technical examination, threat analysis, and English reporting. The site provided English use in context practice and hands-on experience.

2.4 Procedure

Phase 1 – ESP Training in Cybersecurity (For Group 1 Only)

Group 1 members underwent professional ESP training in cybersecurity involving:

- Grammar: Re-emphasis on the right use of the tense, proper use of the passive construction, modal usage, and proper usage of conditionals needed when explaining cyber incidents.
- Syntax: Practice of clear, comprehensible technical sentence structures when reporting and assessing cyber-attacks.
- Vocabulary: Specialist vocabulary concerning terms of cybersecurity e.g., "phishing," "SQL injection," and "bypassing firewalls" and when properly to use each in each setting.
- Derivational Morphology Word Creation: Derivational morphology exercises (e.g., "encrypt/encryption/encrypted") and compound word creation (e.g., "malware analysis").
- Contextual Understanding: Contextual understanding of cybersecurity report language and accurate threat description comprehension based on context.

Group 2 maintained their normal coursework and was subjected to generic English alone without any specific emphasis on cybersecurity English.

Phase 2 – Threat Analysis and Reporting Task (Both Groups)

All the participants had to learn a collection of cyber threat examples presented through TryHackMe and written documents. They had to:

- Know the threat.
- Identify technical features (e.g., type of attack, impacted systems, indicators of compromise).
- Create a comprehensive technical report in English of the threat and suggested mitigations.

2.5 Data Collection

The information was gathered using the following instruments:

- **Language Proficiency Test:** A test designed to measure grammar, syntax, vocabulary, and word formation in the context of cybersecurity. This test was collaboratively developed by ESP specialists and cybersecurity experts to ensure its content validity and technical relevance. It was pilot tested on a representative sample to confirm clarity, reliability, and validity.
- **Threat Report Assessment:** Student reports were assessed on clarity, accuracy, language syntax, and the application of cybersecurity terms. The rubric used for assessment was developed based on established academic and professional criteria and reviewed by subject matter experts to ensure its validity and consistency. The rubric focused on coherence, accurate use of technical terminology, and correctness of cybersecurity concepts.
- **Survey Questionnaire:** Evaluated the students' attitudes towards the relevance of ESP teaching in threat analysis and technical writing.
- **TryHackMe Interaction Observation:** Observed students interacting with the platform, noting their language usage and analytical mind in reporting tasks.

Sample items from the Language Proficiency Test and the detailed Threat Report Assessment rubric are provided in Appendix A to support replicability.

2.6 Data Analysis

Analysis was directed towards both the qualitative and quantitative fronts:

- **Comparison of Performance:** Group 1 and Group 2 were compared for differences in report quality, use of language, and accuracy of threat analysis.
- **ESP Element Assessment:** Grammar, syntax, vocabulary, and word formation were assessed and compared for all the participants.
- **Contextual Understanding:** Assessed the ability of students to read and act on contextual cues in cybersecurity situations.
- **Gender-Based Differences:** Noted and observed trends with regards to risk assessment as well as stylistic differences in documentation between male and female respondents.

2.7 Ethical Considerations

This research adhered to the ethical standards for research involving human participants. Informed consent was obtained from all participants prior to their involvement in the study. Anonymity and confidentiality of the responses were strictly maintained. The study was conducted in compliance with institutional ethical guidelines and was approved by the ethics committee at University Metropolitan Tirana.

3. Results

This section presents the results of the quasi-experimental mixed-methods design used to assess the effect of ESP training on cybersecurity threat analysis and detection. The outcomes are based on quantitative and qualitative data collected from the two participant groups at University Metropolitan Tirana: Group 1, consisting of ESP-trained software engineering students, and Group 2, consisting of non-ESP-trained computer engineering students.

3.1 Comparison of Language Proficiency and Threat Report Quality

3.1.1 Language Proficiency Test

The test scores from the language proficiency test indicated a clear superiority of the ESP-trained group (Group 1) compared to the non-ESP-trained group (Group 2). Group 1 excelled over Group 2 in every category examined: grammar, syntax, vocabulary, and word formation. In particular, the ESP-trained students demonstrated far superior facility with specialized cybersecurity vocabulary (e.g., terminology such as "SQL injection," "phishing," and "firewall bypassing") necessary for effective threat analysis and reporting.

Table 1. Language Proficiency Test Scores (University Metropolitan Tirana, 2024)

Evaluation Criteria	Group 1 (ESP-Trained)	Group 2 (Non-ESP-Trained)
Grammar and Syntax	4.8/5	3.4/5
Vocabulary Usage	4.9/5	3.6/5
Word Formation and Derivations	4.7/5	3.3/5

3.1.2 Threat Report Evaluation

Group 1's reports exhibited greater organization, clarity, and accuracy than Group 2's reports. ESP-trained students used proper grammar, technical terms, and rational report formats to their advantage and hence wrote cohesive, readable reports. Non-ESP-trained students experienced difficulties in both clarity and accuracy, and most of them used technical terms wrongly or made use of uninterpretable sentence structures.

Table 2. Report Assessment Scores (University Metropolitan Tirana, 2024)

Evaluation Criteria	Group 1 (ESP-Trained)	Group 2 (Non-ESP-Trained)
Report Clarity	4.6/5	3.5/5
Technical Accuracy	4.7/5	3.3/5
Sentence Structure and Syntax	4.8/5	3.4/5

3.2 Cybersecurity Threat Identification and Analysis

3.2.1 Threat Identification and Analysis

In terms of identification and analysis of cybersecurity threats, Group 1 had greater ability to pick out the outstanding features of the cyber threats, such as the type of attack, affected systems, and indicators of compromise. They also provided more accurate and actionable mitigation in their reports. Group 2, while capable of finding some dimensions of the threats, had less thorough analysis and less understanding of the overall implications of the attacks. (University Metropolitan Tirana, 2024)

Key Findings:

- 85% of the students in Group 1 correctly identified technical details such as indicators of compromise and attack types.
- Only 58% of the students in Group 2 demonstrated the same proficiency in threat identification.
- Group 1 students demonstrated more ability for providing contextually relevant mitigation strategies.

3.3 Survey Questionnaire: Student Perceptions of ESP Training

A survey was conducted to capture students' perceptions of their experience using both learning methods. 100 students from the Faculty of Software Engineering, University Metropolitan Tirana, participated in the survey. 50 students had received English for Specific Purposes (ESP) training, and 50 had not. Among the respondents, 49% identified as female and 51% as male. (University Metropolitan Tirana, 2024)

They were asked their opinion about: "How important is English proficiency for understanding and analyzing cybersecurity threats?" The result is shown in Fig. 1, below. The number of respondents who voted this aspect as very important is around 32% (32 students), while those who considered it important account for 51% (51 students). 12% (12 students) considered it as neutral, 4% (4 students) as somewhat important, and 1% (1 student) as not important. This reveals that the larger number of the participants confirmed the dominant role of English proficiency in translating and investigating cybersecurity threats.

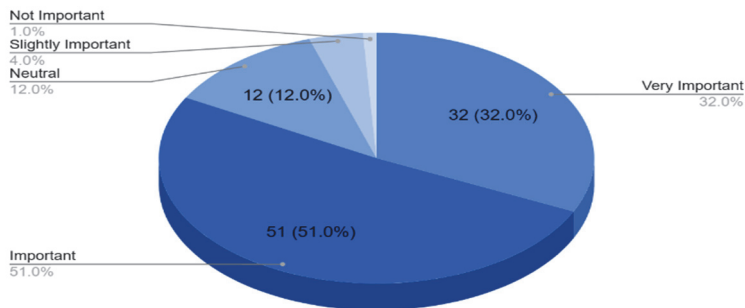


Figure 1. Importance of English proficiency in analyzing and understanding cybersecurity threats.

The previous question is followed by: "How confident are you in your ability to identify technical features (e.g., type of attack, affected systems, indicators of compromise) in a cybersecurity scenario?" The results are shown in Fig. 2, below. 33% (33 students) of the students responded that they were very confident, and 34% (34 students) said they were confident. 27% (27 students) were neutral in confidence level, 5% (5 students) described themselves as slightly confident, and 1% (1 student) said they were not confident. It indicates that the biggest majority of students are confident or highly confident regarding identification of technical features in the event of a cybersecurity scenario and demonstrates the impact of the training in a positive way and highlights how technical linguistic proficiency proves beneficial when it comes to cybersecurity careers.

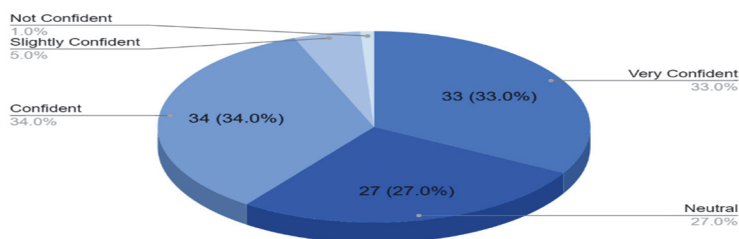


Figure 2. Confidence in identifying technical features in cybersecurity scenarios.

Another question investigated: "How comfortable are you writing a technical report in English about a cybersecurity threat?" Results are shown in Fig. 4, below. 35% (35 students) of the participants indicated they were very comfortable, and 33% (33 students) indicated they were comfortable. 16% (16 students) were neutral, 13% (13 students) indicated they were a little comfortable, and 3% (3 students) indicated they were not comfortable at all. These results show that the majority of students are comfortable or very comfortable writing technical reports in English, which supports the effective influence of ESP training in promoting the technical writing skills of the students in the cybersecurity context.

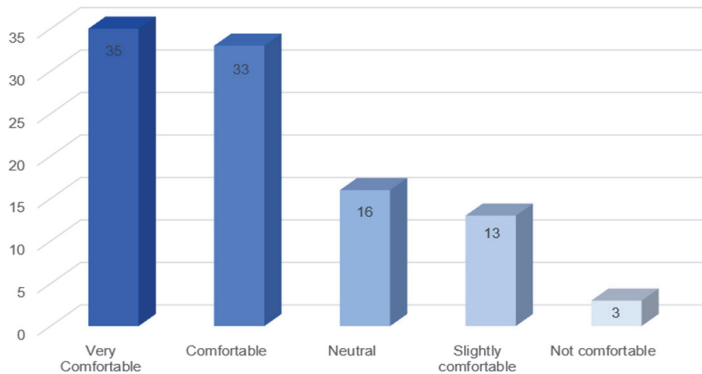


Figure 3. Comfort level in writing technical reports on cybersecurity threats in English.

Students were also asked: "Do you think that having specialized English training (ESP) would improve your ability to understand and use cybersecurity-related terminology?" The results are shown in Fig. 5, below. Of the students, 26% (26 students) strongly agreed, 48% (48 students) agreed, 18% (18 students) were neutral, 7% (7 students) disagreed, and 1% (1 student) strongly disagreed. These responses demonstrate a widespread belief on the part of the majority of students that specialized English teaching (ESP) would significantly enhance their knowledge and application of cybersecurity-specific terminology.

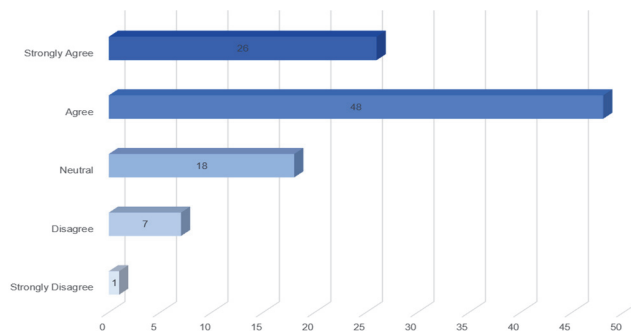


Figure 4. Perceived impact of specialized English (ESP) training on understanding and using cybersecurity-related terminology.

In relation to the effectiveness of ESP training, students were asked: "How much do you think ESP training can help you in writing clear and accurate technical reports in cybersecurity?" The answers

are represented in Fig. 6, below. Of the participants, 30% (30 students) thought that it would be very helpful, 41% (41 students) thought it would be very helpful, 20% (20 students) thought it was quite helpful, 7% (7 students) thought it would be little helpful, and 2% (2 students) thought that it would not be helpful. These findings reveal that the majority of students view ESP training as a helpful tool in enhancing their ability to produce good and correct technical reports in cybersecurity with high positive attitudes towards its impact.

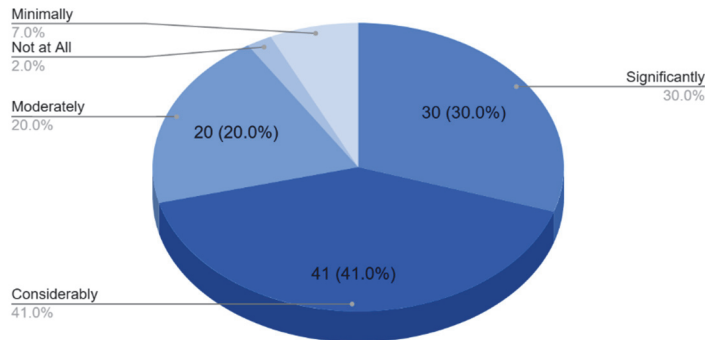


Figure 5. Perceived impact of ESP training on writing clear and accurate technical reports in cybersecurity.

When asked: "In your opinion, how important is understanding grammar and syntax in English for creating precise and coherent cybersecurity reports?" results, as discussed in Fig. 7, show that it was very significant for 29% (29 students), it was significant for 38% (38 students), 18% (18 students) of them were indifferent, 14% (14 students) regarded it as quite significant, while 1% (1 student) did not regard it as significant. These results mean that the majority of the students comprehend the necessity of grammar and syntax in constructing proper and well-structured cybersecurity reports.

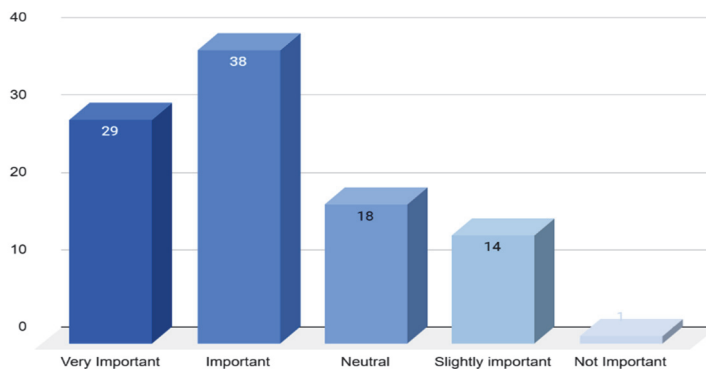


Figure 6. Perceived importance of grammar and syntax in creating precise and coherent cybersecurity reports.

Students were also asked: "Do you believe ESP training would enhance your ability to analyze and report on cybersecurity threats in a professional environment?" The result, as shown in Fig. 8, is that 34% (34 students) strongly agreed, 37% (37 students) agreed, 22% (22 students) were neutral, 6% (6

students) disagreed, and 1% (1 student) strongly disagreed. These findings show that the majority of students believe that ESP training would benefit them significantly in reporting and analyzing cybersecurity threats professionally, and this reflects the perceived significance of certain English training in the field of cybersecurity.

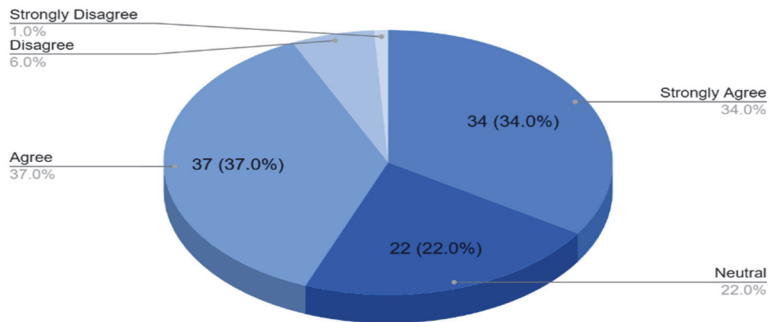


Figure 7. Perceived impact of ESP training on analyzing and reporting cybersecurity threats in a professional environment.

Another significant question was: "How confident are you in using English to describe complex cybersecurity events (e.g., data breaches, malware attacks)?" The outcome, illustrated in Fig. 9, illustrates that 31% (31 students) were very confident, 37% (37 students) were confident, 16% (16 students) were neutral, 15% (15 students) were slightly confident, and 1% (1 student) was not confident. These results imply that the majority of the students were confident in articulating complex cybersecurity incidents in English with a large portion of them being highly confident.

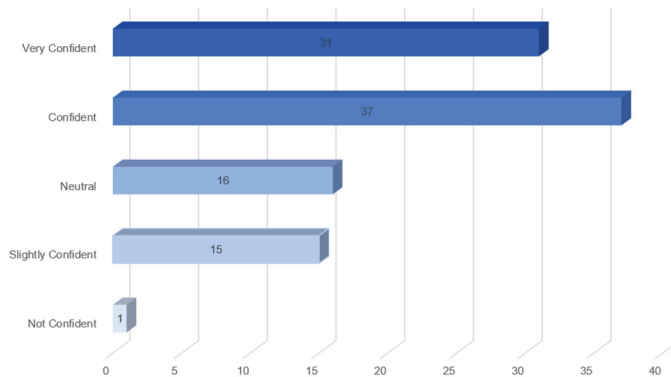


Figure 8. Confidence in using English to describe complex cybersecurity events.

Students were asked: "Would ESP training be beneficial for your career in cybersecurity, particularly in technical communication and threat analysis?" The results, as shown in Fig. 10, are that 25% (25 students) strongly agreed, 46% (46 students) agreed, 23% (23 students) were neutral, 5% (5 students) disagreed, and 1% (1 student) strongly disagreed. These results reveal that the majority of the students recognize the potential usefulness of ESP training for their future professional lives in cybersecurity, particularly for technical communication and threat analysis skills.

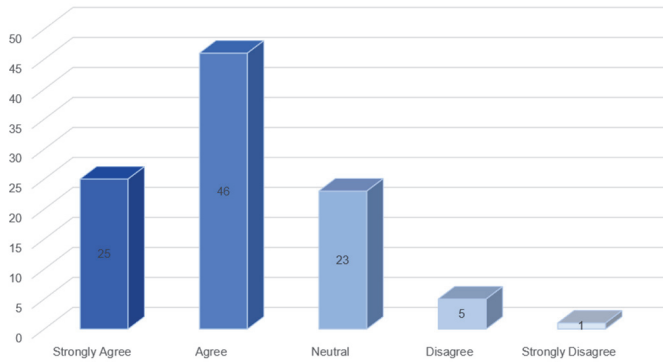


Figure 9. Perceived benefits of ESP training for a career in cybersecurity.

These responses were concluded with this question: "In your experience, how effective are digital platforms (e.g., TryHackMe) in enhancing your cybersecurity skills in conjunction with ESP training?" Fig. 11, below, shows that 30% (30 students) rated the platforms very effective, 37% (37 students) rated them effective, 20% (20 students) were neutral, 10% (10 students) rated them slightly effective, and 3% (3 students) rated them not effective. These results demonstrate that the majority of the students affirm that the incorporation of ESP training with the web-based tools significantly enhances their cybersecurity skill, thereby indicating the effectiveness of the method in providing effective cybersecurity training.

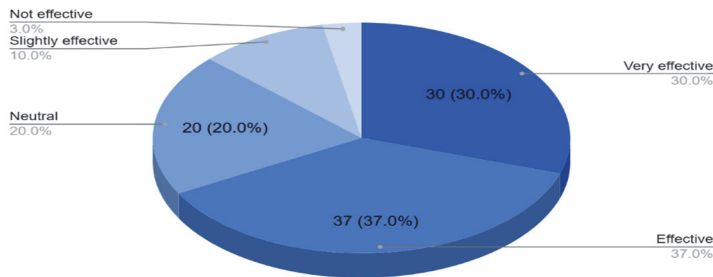


Figure 10. Effectiveness of digital platforms (e.g., TryHackMe) in enhancing cybersecurity skills alongside ESP training.

3.4 TryHackMe Interaction Observations

Observational data from the TryHackMe exercises showed that students who had undergone ESP training were more competent in the use of the platform and in interacting with the simulated cyber-attacks. Group 1 students were more engaged with the report assignments, where the students provided higher levels of analysis and concise wording. Group 2 students were able to perform tasks but lacked crucial details within the reports or failed to provide answers to tasks using proper technical terminology. (University Metropolitan Tirana, 2024)

Key Observations:

- 92% of students in Group 1 were intensely involved with the TryHackMe platform, in contrast to 68% of students in Group 2.
- Group 1 students provided complete reports based on best practices for technical writing, grammar, and spelling, while Group 2 students often left out key threat characteristics or presented confusing text.

3.5 Instructors Evaluation

Instructors evaluations were conducted by an experienced panel of three instructors representing the University Metropolitan Tirana who were engaged through the research duration. The three instructors, all with technical writing, cybersecurity, and English for Specific Purposes (ESP) expertise, comprehensively evaluated the performance of students. They awarded the ESP-trained students a notable improvement in terms of technical writing and cybersecurity analysis compared to non-ESP-trained students.

The instructors highlighted the effectiveness of the ESP training in equipping students with necessary language skills to function in cybersecurity settings. Specifically, they noted that ESP-trained students excelled considerably in producing clear, concise, and technically accurate reports, showing their enhanced ability to utilize specialized cybersecurity vocabulary, appropriate grammar forms, and analytic thinking in real-world contexts. Feedback highlighted the critical role of ESP training to close the gap between technical knowledge and language ability in the field of cybersecurity.

Table 3. Instructor Ratings (University Metropolitan Tirana, 2024)

Evaluation Criteria	Group 1 (ESP-Trained)	Group 2 (Non-ESP-Trained)
Grammar and Syntax	4.8/5	3.5/5
Technical Vocabulary Usage	4.9/5	3.7/5
Analytical Thinking in Threat Reporting	4.7/5	3.3/5
Overall Professional Performance	4.9/5	3.6/5

The findings of this study highlight the importance of ESP training in enhancing not only language skills but also technical know-how required in cybersecurity careers. The performance difference between ESP-trained and non-ESP-trained students clearly shows that special language training is essential to technical success in such fields as cybersecurity. Integration of ESP programs within technical studies can be a good measure to improve professional communication skills as well as analytical skills. Future research can explore the lasting impact of ESP training on career achievement in the field of cybersecurity, and extend it to other technical fields.

3.6 Mastery of ESP Components

Intensive instruction in derivational morphology together with compound formation instruction and contextual learning produced quantifiable results.

- Group 1 employed compound terms precisely while confirming their knowledge of phenotypical changes (encrypt → encrypted → encryption) in terminology. Warning indicators in scenario-driven prompts became easier to detect for this group while they demonstrated professional expertise through proper mitigation recommendations using linguistically correct terminology.
- Technical success in both field-specific morphology and syntax enables students to learn organizational discourse techniques necessary for their transition from education into professional employment.

3.7 Contextual Understanding in Cybersecurity Situations

During ambiguous threat assessments Group 1 students proven successful at analyzing the situation and determining weak points before creating cohesive threat assessment models.

Example:

- Group 1 students examined a simulated log entry's suspicious outbound activity and detected data theft potential which led to the development of network protection methods utilizing appropriate technical language.

The educational approach helps students reach a proficient vocabulary level while they develop skills to analyze and solve problems in contextual situations which they will encounter in high-security cyber operations.

3.8 Gender-Based Trends in Technical Communication

Stylistic patterns emerged between male and female participants in their performance outcomes though the statistical results were not significant.

- The male students displayed stronger assertiveness when identifying risks and taking decisions by choosing direct action terminology during their reports.
- Female students used structured language formats in their reports because they focused on producing detailed information with careful attention to narrative details.

The observed communication patterns create opportunities to develop adjustable ESP teaching approaches which support different student expression techniques and sustain technological precision.

3.9 Emerging Role of Digital Platforms

The TryHackMe platform supplied essential real-time learning opportunities regarding language employment alongside analytical process development. The members of Group 1 combined their tasks with technical communication activities for problem-solving but Group 2 centered their work on task fulfillment instead of language perfection.

The combination of virtual cybersecurity labs and ESP gives students rich environments to simultaneously learn skills and language complexity which shapes a promising model for classroom development.

4. Discussion

Quasi-experimental comparison of ESP-trained software engineering students with non-ESP-trained computer engineering students indicates the crucial intervention of domain-specific language skills in bridging the gap between technical expertise and effective communication—a prelude to cybersecurity success. The findings are congruent with the established ESP theory that contends language instruction intended for workplace contexts empowers learners with the syntactic, lexical, and pragmatic means needed for task-focused performance (Hutchinson & Waters, 1987; Dudley-Evans & St. John, 1998). For instance, ESP-trained participants demonstrated improved control over cybersecurity terms (e.g., "zero-day exploit," "privilege escalation") and syntactic structures (e.g., passive voice for reporting incidents), enabling precise articulation of threat patterns and mitigation strategies.

Through modeling of real-world scenarios such as analyzing phishing campaigns and penetration testing, the platform forced students to employ specialized terms and grammar on practical, realistic assignments—like generating incident reports that require simplicity (e.g., "The payload was transmitted by means of an infected attachment"). This is in line with research

encouraging interactive, technology-enhanced language learning (TELL) tools to facilitate both linguistic and technical capabilities (Stockwell & Hubbard, 2013). The game-like nature of TryHackMe also solved the usual ESP course challenge: keeping the learners motivated in the absence of exciting technical writing exercises (Chou, 2016).

Gendered differences in communication style—male interviewees' assertiveness in risk assessment compared to female interviewees' care for documentation—are part of broader sociolinguistic trends in STEM education (Carli, 2001).

However, the quasi-experimental design introduces limitations. Existing disciplinary distinction between software and computer engineering students—uneven exposure to cybersecurity topics—is most likely to have confounded ESP training effects with inherent curricular bias. Second, self-reported survey measures are vulnerable to social desirability bias because participants will tend to overreport perceived ESP training benefits. Future studies should employ randomized controlled trials and longitudinal follow-up to control for ESP's effect and assess skill retention in the workplace.

5. Conclusion

In conclusion, this study verifies that ESP training, if complemented by interactive learning platforms like TryHackMe, significantly enhances students' ability to detect, analyze, and explain cyber threats. Through the elimination of the linguistic-technical gap, ESP empowers learners to decode complex threat warnings (e.g., IOC models), accurately explain mitigation techniques, and collaborate productively within multilingual teams—a critical ability in a world of globalized cyber threats. Those same gender differences in communication style are also why a diverse cybersecurity team is of such value, in which assertive risk assessment and detailed documentation are equally essential.

For policymakers and educators, these findings justify the formal incorporation of ESP modules into cybersecurity education, particularly in non-Anglophone contexts where technical English proficiency remains a barrier. Online learning environments like TryHackMe offer scalable, cost-effective means to simulate real-world language demands, although institutions must address disparities in resources in order to enable equal access. Subsequent research should explore the longitudinal impact of ESP training in the workplace and evaluate adaptive learning models from AI tools (e.g., LLMs) to personalize language training.

Overall, as threats over the Internet grow more sophisticated, the integration of ESP and practical technical training will be worth its weight in gold in creating an employee base capable of defending cyber infrastructures while also being able to explain the defenses effectively. Linguistic and technical skills on both sides is not only good—world cybersecurity resilience requires it.

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Appendix A:

A.1 Language Proficiency Test Sample Items

The language proficiency test was collaboratively designed by ESP and cybersecurity specialists to evaluate grammar, syntax, vocabulary, and word formation within the cybersecurity context. It was pilot tested for clarity and validity. Sample items include:

- Grammar and Syntax

Choose the correct sentence:

- a) The phishing attack was detected by the security team.
- b) The phishing attack detect by the security team.
- c) The phishing attack detecting by the security team.

- Specialized Vocabulary Matching

Match the terms with definitions:

Table 4. Sample Task: Match the terms with definitions

Term	Definition
SQL Injection	Unauthorized insertion of malicious code into a database
Malware	Malicious software designed to damage or disrupt systems
Firewall	A system that monitors and controls network traffic

- Word Formation and Derivations

Choose the correct noun form of “encrypt”:

- a) encrypted
- b) encryption
- c) encrypting

A.2 Threat Report Assessment Rubric

The rubric was developed based on academic and professional standards, validated by subject matter experts, and used to assess students' cybersecurity threat reports on the following criteria:

Table 5. Sample Threat Report Assessment Rubric

Criterion	Description	Scoring (1=Poor to 5=Excellent)
Clarity and Organization	Logical flow, coherence, and readability	
Technical Accuracy	Correct identification of threats, impacted systems, and indicators of compromise	
Grammar and Syntax	Correct use of grammar, including passive voice and modal verbs	
Use of Specialized Vocabulary	Precise and contextual use of cybersecurity terminology	
Analytical Depth	Depth of threat analysis and relevance of mitigation suggestions	

A.3 Survey Questionnaire Excerpts

Sample items used to evaluate students' attitudes and perceptions regarding ESP training's relevance and effectiveness:

- How important is English proficiency for understanding and analyzing cybersecurity threats?

(Scale: Very Important – Not Important)

- How confident are you in identifying technical features (attack type, affected systems, indicators)?
(Scale: Very Confident – Not Confident)
- Do you believe ESP training improves your ability to write technical cybersecurity reports?
(Scale: Strongly Agree – Strongly Disagree)
- How effective are digital platforms (e.g., TryHackMe) in enhancing your cybersecurity skills alongside ESP training?
(Scale: Very Effective – Not Effective)

A.4 TryHackMe Interaction Observation Protocol

Observers monitored participants during practical threat analysis tasks on TryHackMe, focusing on:

- Engagement level with the platform activities
- Accuracy and technical detail in students' reports
- Use of specialized cybersecurity vocabulary
- Clarity and coherence of explanations
- Analytical approach in threat identification and mitigation

A.5 Data Collection Summary

- Participants: 50 ESP-trained software engineering students and 50 non-ESP-trained computer engineering students.
- Data Sources: Language proficiency tests, threat report assessments, survey questionnaires, and observation notes from TryHackMe sessions.
- Validation: Instruments were pilot tested and reviewed by ESP and cybersecurity experts to ensure content validity, clarity, and reliability.